

Surya Ravi

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EDUCATION

Kingston University <i>Master's Degree in Computer Animation (School of Computer Science & Mathematics)</i> Award: Distinction	Sep. 2023 – Sep. 2024 London, United Kingdom
Kumaraguru College of Technology <i>Bachelor's in Mechatronics Engineering (Affiliated to Anna University, Chennai)</i> GPA: 8.8	Aug. 2019 – May 2023 Coimbatore, Tamil Nadu
PVV Higher Secondary School HSC: 91.4% & SSLC: 94.8%	Apr. 2017 – Apr. 2019 Coimbatore, Tamil Nadu

EXPERIENCE

Orbia <i>Senior Analyst – Threat Exposure Management</i> - Designed, developed and currently maintain 18 end-to-end automations using Tines SOAR covering threat monitoring, alert triage, vulnerability management workflows and false positive reclassification, improving response efficiency, decreasing manual work hours and reducing the number of false positive reports. - Validated comprehensive penetration testing reports, triaging bugs and ensuring accurate documentation of vulnerabilities, proof-of-concepts, and remediation recommendations in Intigriti and HackerOne. - Developed 5 CrowdStrike Falcon Fusion SOAR workflows to automate security response and remediation actions, enabling faster investigations, decision-making and endpoint response while reducing manual analyst intervention. - Executed 2 white-box penetration tests on internal web applications, identifying security gaps and collaborating with engineering teams for remediation. - Developed and optimized security orchestration workflows integrating CrowdStrike, Tenable, ServiceNow, Assetnote and Proofpoint to prioritize vulnerabilities in endpoints and application environments. - Built API-driven security automation workflows using REST APIs, Postman, JSON and Python, enabling integration and processing of security data across disparate security platforms. - Managed the end-to-end vulnerability lifecycle, from identification and risk assessment through remediation tracking, validation/retesting, stakeholder coordination, and compliance reporting. - Performed remediation validation and retesting to verify successful patch implementation and confirm vulnerability resolution in 8 internal applications.	Dec. 2025 – Present Pune, Maharashtra
Alice <i>AI Red Team Researcher – Freelance</i> - Conducted AI red-team and adversarial security testing using Alice, developing attack scenarios to evaluate model behavior, prompt-injection resistance, guardrails, and security boundaries. - Built and iterated on automated security-testing harnesses for adversarial prompt execution, response capture, reproducibility, and regression testing across attack scenarios. - Identified and validated guardrail bypasses and unintended model behaviors, documenting reproducible attack paths, security impact, and defensive improvements.	2026 Remote
HackerOne <i>Bug Bounty Hunter</i> - Hunted for web and API vulnerabilities in public HackerOne programs, focusing on access control, authentication, rate limiting and exposed staging infrastructure. - Reported 8 High vulnerabilities and 3 Medium vulnerabilities with auth, API and injection vulnerabilities, documenting reproduction steps, explaining security impact and remediation guidance. - Used Burp Suite for DAST and developed automated pipelines for SAST reconnaissance, request tampering and validation of findings, staying within program scope and disclosure rules.	Feb. 2025 – Nov. 2025 Remote
Hackup Technology <i>Cybersecurity Analyst Intern</i> - Conducted hands-on web application security testing using Burp Suite, analyzing application behavior, identifying security weaknesses, and developing practical understanding of vulnerability discovery and validation. - Performed network traffic analysis using Wireshark, examining protocols, packet flows, and network communications to strengthen understanding of security monitoring and detection of anomalous activity. - Gained practical exposure to SOC operations, threat monitoring, vulnerability management, and security assessments, developing foundational experience in identifying, analyzing, and responding to security issues. - Delivered technical seminars across 3 colleges covering Linux fundamentals, command-line operations, and cybersecurity concepts, translating technical security topics into practical guidance for students. - Developed practical knowledge of web application vulnerabilities aligned with PortSwigger Web Security Academy, including SQL Injection, IDOR, API security flaws, NoSQL Injection, Reflected and Stored Cross-Site Scripting, Authentication and Access Control issues, and other OWASP-aligned vulnerabilities, with hands-on focus on exploitation, validation, and remediation techniques.	Aug. 2025 – Nov. 2025 Coimbatore, Tamil Nadu

PROJECTS

- arc-ai** – PoC-generating security agent
- LangGraph + Temporal agent that finds vulnerabilities from source, proves each with a working exploit in an egress-locked sandbox and reports only verified findings; proves 3/3 planted vulnerabilities (SQLi, SSRF, IDOR) and discards unprovable hypotheses, for zero unverified alerts.
- taintline** – Static taint analyzer
- Go + tree-sitter analyzer tracing user input to SQL injection, command injection and path-traversal sinks in Python and JavaScript, with SARIF output for GitHub Code Scanning; detects 9/9 seeded flaws with 0 false positives on safe twins, backed by 42 engine edge-case tests.
- PPx** – Prompt Pattern Index
- LangGraph pipeline that classifies prompts across 4 prompt-injection pattern families (prompt extraction, instruction override, data exposure, malicious patterns) and produces a risk score, severity rating and security report.
- Attack Surface Mapper**
- Application-understanding engine that maps a codebase in 4 discovery stages (framework, routes, authentication, external services) before vulnerability hunting; surfaced 37 routes, 4 admin routes and 2 external services on a sample Express app.

WRITEUPS

- *Alice LLM Security* – LLM-focused security research through prompt injection, instruction-hierarchy manipulation, adversarial prompting, model steering, and output-format coercion while solving an AI security CTF. Awarded Best Technical Write-up. [Read](#)
- *QKD* – Research on Quantum Key Distribution and its cryptographic foundations, exploring quantum principles, secure key exchange, eavesdropping detection, and the role of QKD in post-quantum cybersecurity. [Read](#)

SKILLS

Primary Skills: SOAR, Incident Response, MITRE ATT&CK, Threat Intelligence, API Security, OWASP Top 10, Cryptography, Vulnerability Management, Python, PowerShell, C

Secondary Skills: Bash, Linux, Penetration Testing, Quantum Cryptography, CI/CD, LLM Security, Adversarial Testing, Relational and Non-relational database

Stack: Tines, Burp Suite, LangGraph, Postman, Git, GitHub, VS Code and Docker

Certifications: GCPC, Tines Certified Professional, Postman API Developer Associate, CrowdStrike Falcon Fusion SOAR and LangGraph Foundations

Soft Skills: Problem-Solving, Fast-paced Learning and Cross-functional collaboration

Hobbies: Guitar, Surfing, Sudoku, Rubik’s Cubes, Chess, Reading

HONOURS & AWARDS

Name	Placing	Type	Scope	Awarder	Year
Alice in Wonderland LLM CTF	Top 5	AI Security CTF	Global	Alice	2026
Kriya CTF (Jeopardy)	1 st Place	Security CTF	IN	PSG Institute of Technology	2020
IISF Hackathon	1 st Place	Hackathon	Global	Ministry of Sci. & Tech.	2020
Robo Sumo	1 st Place	Robotics	IN	PSG Institute of Technology	2021
Outstanding Student Award	Academic & Inter-College Event Excellence	Academic Award	IN	Kumaraguru College of Tech.	2023
Certificate of Recognition	Academic Merit	Academic Award	IN	Kumaraguru College of Tech.	2023
Mahatma Gandhi Merit Scholarship	Top 10% of Institution	Scholarship	IN	Kumaraguru College of Tech.	2019, 2020